02 - 02 - 2025

Agenda —

- Yolo v1 →
- Yolo v2 →
- → v3
- → v4
- → v5

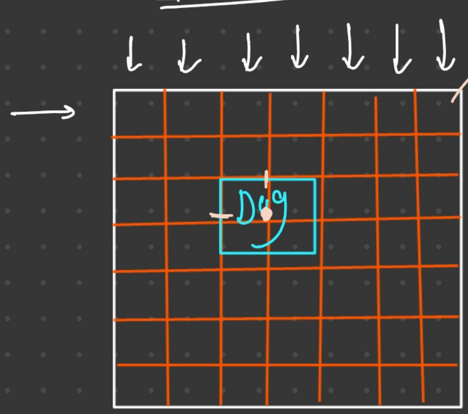
→ project Discussion

448×448

7×7

$$\begin{array}{r} \boxed{64} \\ 7\overline{)448} \\ 42 \\ \hline 28 \\ 28 \\ \hline 0 \end{array}$$

Yolo v1



64 x 64 pixels

(i) $\boxed{7 \times 7}$

Dog, person ← C → 2

448×448

→ 64

txt

class	x	y	w	h
	x_c	y_c	w	h

h

w

- Each grid cell predict 2 bbox

$$[\,[x, y, w, h, obj_{conf}], [x, y, w, h, obj_{conf}]\,],$$

[Dog person]

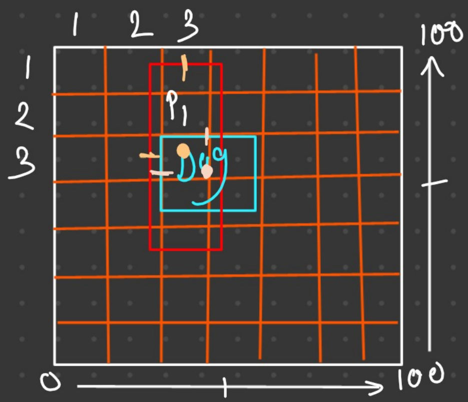
$$\underbrace{5 \times 2 + C}_{\boxed{10 + C}} \times (7 \times 7)$$

$49 \times [10 + C]$

$49 \times [10 + 2]$

49×12 →

→



(3,3) → Dog

P1 → xc, yc
Dog → xc, yc
(3,3)

(3,3) → [xc, yc, w, h] ① 30, 60, 10, 20, 0.9

[xc, yc, w, h] ② 35, 70, 20, 10, 0.5

[0.8 0.7]

Dog person

Box 1: $c = 0.9$

$p(Dog) * conf = 0.8 \times 0.9 = 0.72$

Box 2: $c = 0.5$

: $p(Dog) * 0.5 \rightarrow 0.8 \times 0.5 = 0.40$

$0.72 > 0.35 \longrightarrow$ $\boxed{\text{Box 1}}$

Yolo1 → cannot predict multiple objects in a grid.

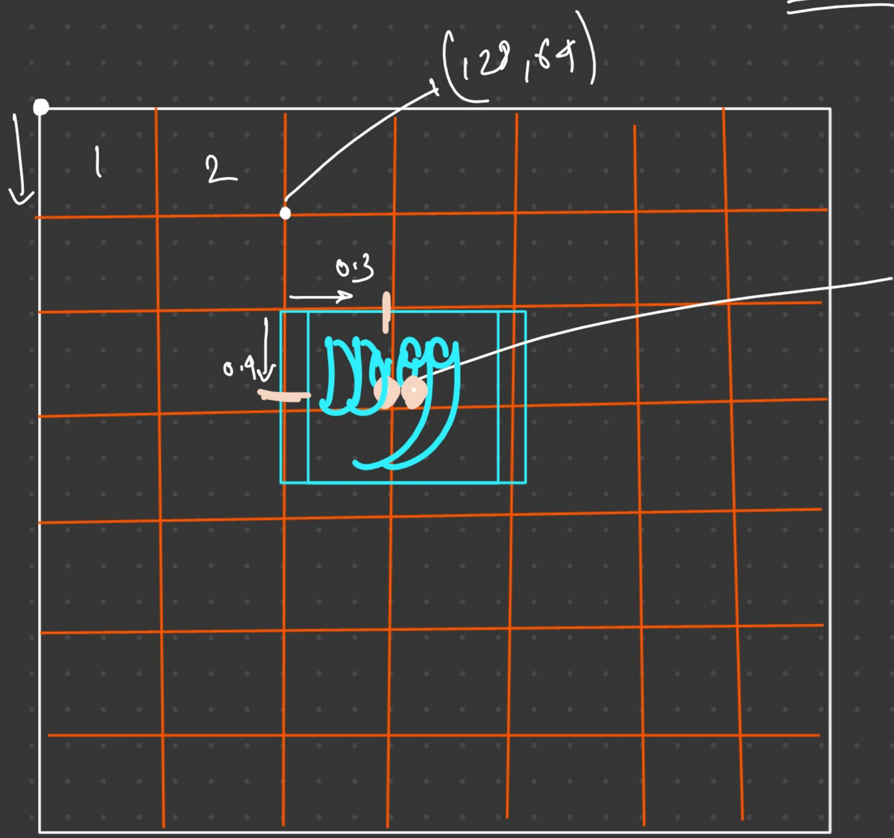
$\rightarrow \quad x_c, y_c, w, h \quad \rightarrow$

range → 0 − 1

predicted → x, y → 0.3, 0.4 → relative to your grid cell

w, h → 0.5, 0.6

→ <u>relative</u> to whole image



0.3, 0.4

$x = 64 \times 2 \rightarrow \boxed{128}$

$y = 64 \times 1 = \boxed{64}$

(3, 2)

(3, 3)

(3, 3)

$\boxed{(4, 3)}$

predicted → x, y → 0.3, 0.4

64x3 →

w, h → 0.5, 0.6

centre → $x = (\boxed{0.3} + \boxed{3}) \times \boxed{64} \rightarrow x_c \rightarrow \boxed{211 px}$

$= (0.4 + \boxed{3}) \times \boxed{64} \rightarrow y_c \rightarrow \boxed{217.6 px}$

$w = 0.5 \times 448 \rightarrow 224$

$h = 0.6 \times 448 \rightarrow 268$

→ final bbox : 211, 218, 224, 268 → rplot

text

loss

(1) localization loss + (2) confidence loss + (3) classification loss. = final loss → backprop → SGD → Adam

(1) localization loss

MSE

$x, y, w, h \rightarrow GT \rightarrow 0.5, 0.5, 0.4, 0.3$

$x, y, w, h \rightarrow pred \rightarrow 0.3, 0.4, 0.5, 0.6$

$$\left[(0.5-0.3)^2 + (0.5-0.4)^2\right] + \left[\left(0.4^{0.5} - 0.5^{0.5}\right)^2 + \left(0.3^{0.5} - 0.6^{0.5}\right)^2\right]$$

Alternately

$$\boxed{+\left(\sqrt{0.4} - \sqrt{0.5}\right)^2}$$

f_L =

(2) conf loss →

ideal − 1 → object

ideal — 0 → no-object

$$f_{CL} = \sum^{obj} (0.9-1)^2 + \sum^{no.obj} (0.3-0)^2$$

$$= 0.01 + 0.09$$

$$= 0.1$$

$$f_{CCL} = \sum \left(P_c - \hat{P_c}\right)^2$$

$$= (1 - 0.8)^2 = (0.2)^2 = 0.04$$

$$f = f_L + f_C + f_{CCL} = \boxed{0.04 + 0.1 + 0.388}$$

$$=$$

$\downarrow$

BP $\longrightarrow$ GD $\to$ SGD, ADAM, RMS prop

$\boxed{\text{Google LeNet}}$

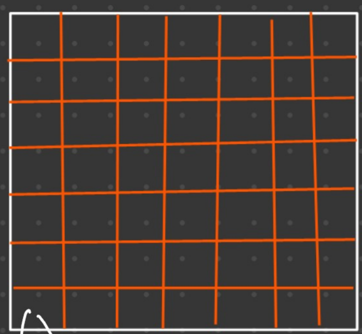
Yolo V2

→ resolution 416 / 448 (mentioned in resource)

→ 13×13 grid

416 / 13 → 32×32

448 / 13 → 34.46 ≈ 35 ≈ 35×35



1 class probabilities

2 bbox, $conf_{obj}$

→ Each grid cell is responsible for predicting objects whose centre fall in the grid cell.

→ Each grid has 5 anchor boxes

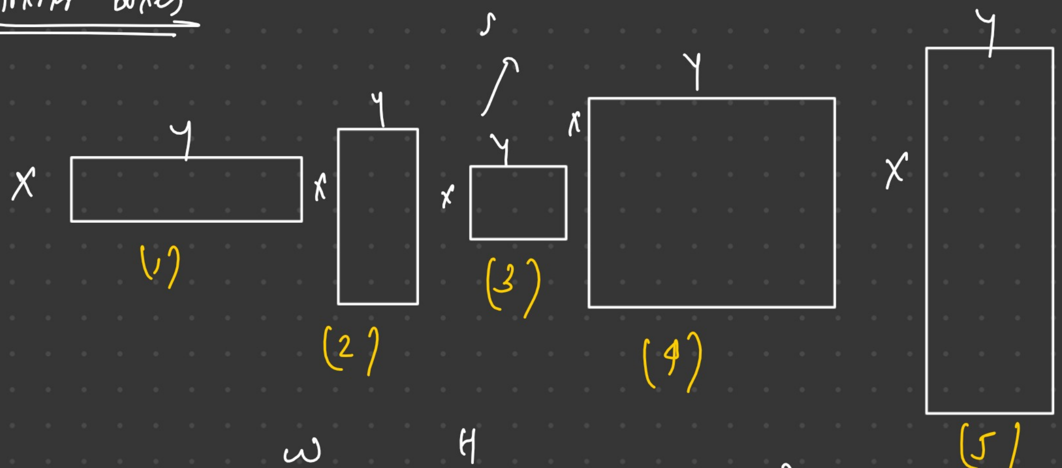
→ for anchor in anchor_boxes:

predict (t_x, t_y, t_w, t_h)

predict (obj)

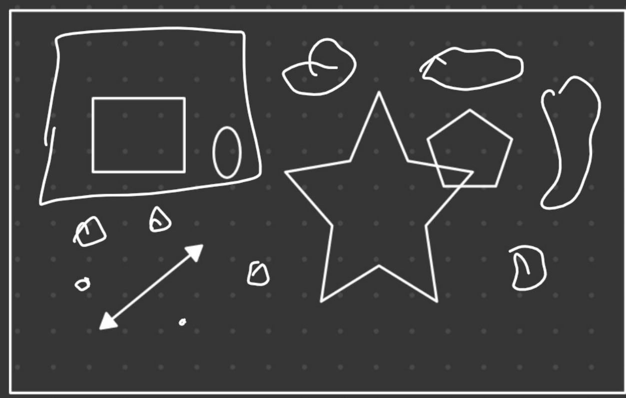
pred (class probabilities)

Anchor boxes



	W	H	
(1) A-1	0.57	0.67	→ square - S
(2) A-2	1.87	2.06	→ Rect - S
(3) A-3	3.33	5.47	→ Rect - m
(4) A-4	7.88	3.52	→ Rect - L
(5) A-5	9.77	9.16	→ square - L

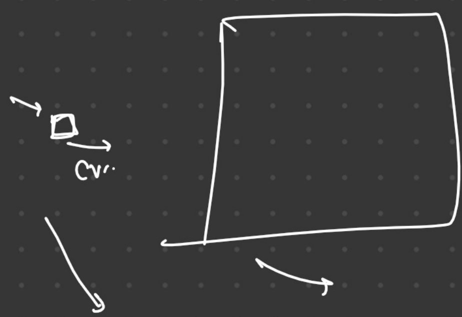
COCO

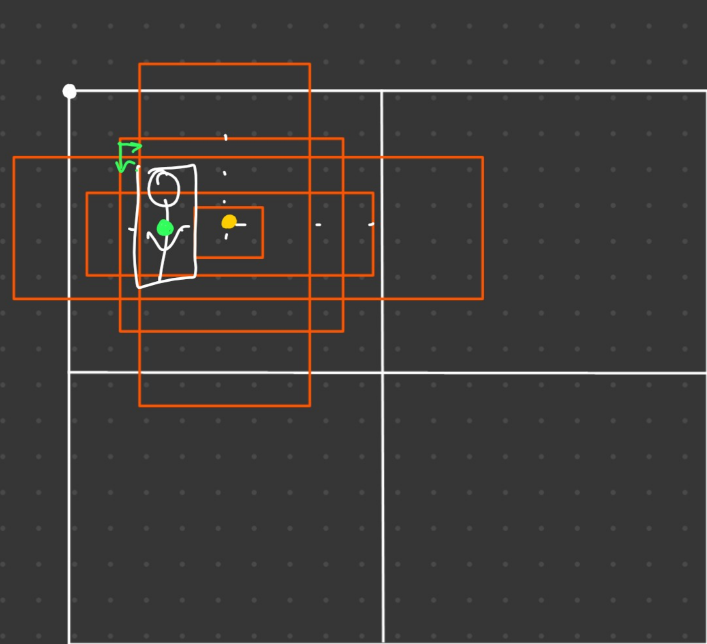


KNN → segment mask

→ Area

largest _ _ small





$[tx, ty, tw, th]$

relative to grid

relative to anchor box

x, y, w, h

Grid →

1 anchor / 1 prediction

$tx,\ ty,\ tw,\ th,\ conf_{obj},\ \overline{Prob}_c$

$$b_x = \sigma(t_x) + c_x$$

$$b_y = \sigma(t_y) + c_y$$

$$b_w = p_w \cdot e^{t_w}$$

$$b_h = p_h \cdot e^{t_h}$$

Anchor box

$$5 \times (4 + 1 + C) = 5 \times (5 + 20) = 5 \times 25 = \underline{\underline{125}}$$

20

13x13 → 13 x 13 x 125 = 21125 o/p

$[t_x, t_y, t_w, t_h], [obj], [[- - - -]]$

	Yolov1	Yolo v2	Yolov3	Yolo v4	Yolo v5
Grid	7x7	13x13	13x13	—	—
Anchor box	No Anchor box	5	3 anchor box per scale (3 scale)	—	—
Multiple scale	No	No	Yes FPN FPN (3 scales)	PANet + FPN	—
loss function	MSE + cross + object	same as v1	Binary cross entropy	CIoU + BCE loss	—
Batch Normali	No	Yes	Yes	Yes	✓
Architecture	GoogleNet	Darknet-19	Darknet-53	CSPDarknet 53	CSP Darknet + Focus
Activation	leaky Relu	leaky Relu	leaky relu	Mish	Swish SiLU

project

(1) Data validation

135 → gdrive → 5 → validate

↓

good bad

↙

gdrive →

(2)

(1) person 1 → $\boxed{\text{car - 1}}$

→ p - 3

— sw - 2

→ person - 1

car - 2

sw - 3

p - 2

c - 2

sw - 1

p = 3

images labels

py →

p-1 → car, bike, road

p-2 → $\boxed{\text{car}}$ $\boxed{\text{bike}}$ → → car bike

p-3 → bike, road →

car → 0, 2, 3, 8

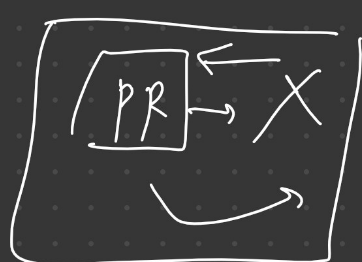
10 - 12

5 - 6

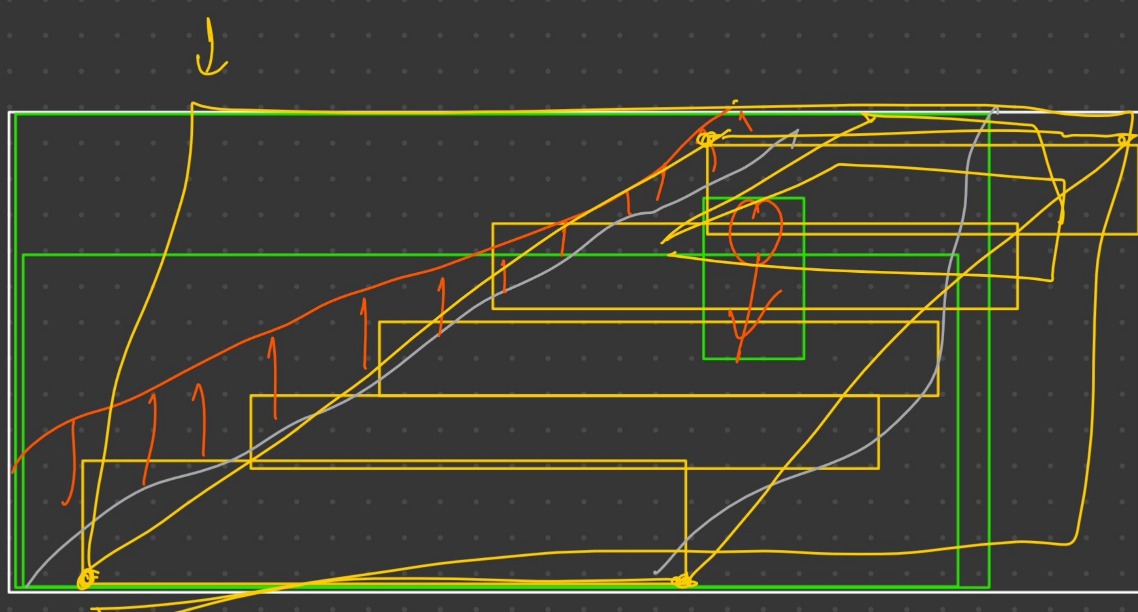
txt

0 1 2 3

partition

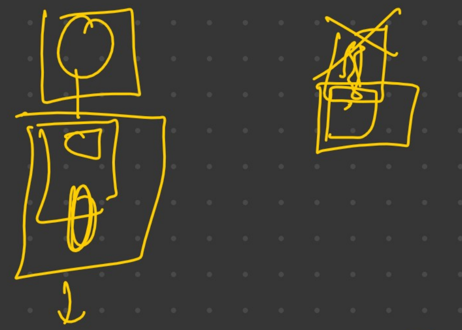


135 → 10



for road:

min x, y



feed →

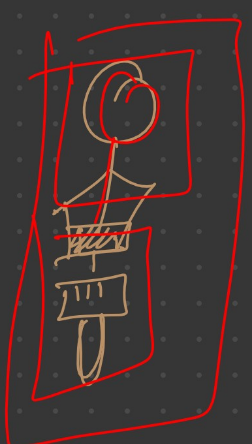
pad → yolo v5 →

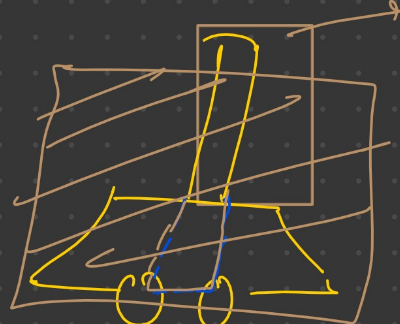
txt → img

img

yolo → pan

txt → img

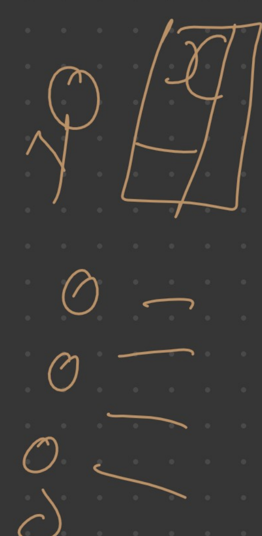




text

features → Annots → CV

features



(1) Data validation

(2) Preprocess → final dataset →

(3) Model train → v5 [v8] →
→ Augm [yolo v8] →
→
→ FP

(4) Backend — Yolo — [detect.py] → [illegible]

→ infer →

[illegible] data

class [Detector] (

→ yolov8

[bbox]

video stream, image, rtsp

CPU 30 →

[fps] → [illegible] [5]

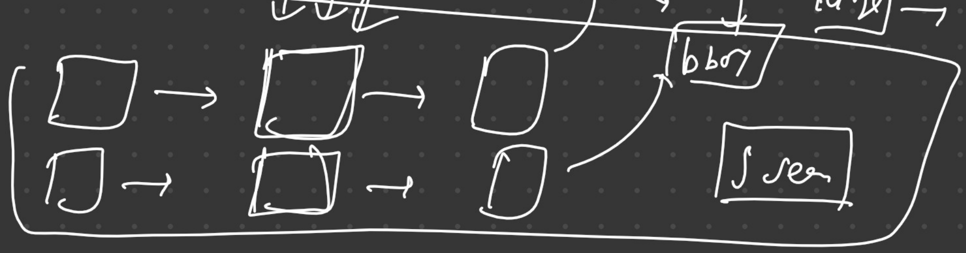
CPU [1]

[stream.py] when [person is coming turn on] → [5]
alert (warn, person coming yes)
way

[Detector]

[bbox] [alarm] → [illegible]

[bbox]



problem statement:

check → if person coming →

→ check if walking on side wall

left → right

Scen 1

Scen 2 → Scen

Scen 3 :

(1) Pre

→ Tr

— Eval

Datable → env

pyth. pn

Cam 1

api (——, ——)

Cam 2

Scenario 1

scer 2

NMJ

venet

c2f

FPN

pnh

a = 2
b = 2
c = a+b